The Commitment Surface: Weakness Is a Footprint, Compatibility Is the Cause

Author. Jawaun Brown, with agent-generated code and drafts under human

direction.

Status. Draft; results embedded per experiment. Format: NeurIPS/ICML style

(no page cap; structured sections, citations, appendix).

Abstract

Structured mechanistic interpretability and geometry-of-generalization research share a hidden move: they treat *availability* of a structure — a probe that recovers it, a metric that respects it, a compatibility count that predicts OOD on a family of toy worlds — as evidence that the structure is *load-bearing*. This paper argues that move is wrong. We reframe the target primitive from "right geometry / right weakness ⇒ generalization" to **commitment-surface survival**: a representation is real for a deployment only if a train-time compatibility intervention with the deployment generator produces a causal effect (patch-CE) at the commitment surface, and that effect survives gauge-fixing and change of commitment. Concretely, we (i) formalize the commitment surface as a triple `(G_dep, C, T)` and prove weakness reduces to Bennett's principle when the probe group matches the deployment generator; (ii) show a strong within-lab discriminator on cyclic modular addition where unweighted weakness and misspecified concern selectors underperform well-specified concern-weighted selection by a decisive margin; (iii) show that neural training with cyclic-orbit augmentation dominates readout selection over trained-with-no-augmentation seeds on OOD accuracy AND patch-cross-entropy, with a wrong-group control at zero; and (iv) run a non-degenerate external-contact sweep on Pythia 70m/160m/410m LoRA-fine-tuned on modular addition, where the compatibility-augmented arm clears OOD while the readout arm hard-kills, closing the P1 external gap our prior work identified. We recover the old-frame positives — cyclic and dihedral 100%-vs-0% weakness sweeps — as the boundary case where the probe group and deployment generator coincide, and interpret the correction chain of autonomous-probing agents as an anti-Goodhart control loop — *detect* → *allocate* → *saturate* → *cool* → *reopen* — whose load-bearing signal is commitment-cooling under intervention-pinned residuals.

1. Introduction

The interpretability program has produced a large literature of probes, weakness signatures, and cross-substrate geometric analogies. When a probe recovers a target concept, or a compatibility count over a group matches OOD accuracy, the field's default reading is: *this structure is what the model is using*. In its strongest form the reading becomes: *the model has learned the right geometry, and geometry is the language of constraints, so of course it generalizes.*

This paper questions that default reading. In our prior program we tested it seriously across ten experiments — cyclic and dihedral modular arithmetic (100%-vs-0% wins for weakness against loss, MDL, flatness, compression), grid cell weakness/topology mediation (near-null), passive→active geometry on real LLMs (7× lift in a paraphrase axis), semantic concern deformation (negative transfer), Suite C world-change re-engagement gates, and external contact via Pythia LoRA on modular arithmetic (P1 hard kill). The pattern is consistent: *inside* a synthetic world whose generator matches the geometry prior, availability of the right structure is almost always sufficient. *Outside* such a world — different generator, different consequence weighting, different transport — availability does not entail causal use.

We name the missing primitive **commitment-surface survival**: a representation is *load-bearing at a commitment surface* iff (i) a train-time compatibility intervention with the deployment generator lifts OOD, (ii) causal patching of the aligned mechanism produces a CE ≥ ε at the commitment target, and (iii) the effect survives gauge-fix and change of commitment. Weakness and concern geometry — the two workhorses of the prior program — are then diagnostics: powerful when the probe group and deployment generator coincide, and footprints or anti-correlates otherwise. The commitment surface is where the discipline is imposed. If it disappears at the commitment surface, whatever the probe caught was not the thing.

Contributions.

`(G_dep, C, T)`; load-bearing structure as CE ≥ ε that survives transport under `T`; probe AUC ⊥ CE without a commitment term (Prop. 1).

concern-weighted weakness (extension mass weighted by consequence) to commitment-pinned intervention (Prop. 2 + Corollary), with cyclic/dihedral results as the aligned-generator special case, not the general law.

well-specified concern beats unweighted and misspec is *worse* than unweighted (misspec at –0.05 vs unweighted, gap = +0.24 for well-specified). E2/E3 show that a neural compatibility-augmented arm dominates a weakness-readout selector on OOD by a decisive gap, with patch-CE aligned. E4 (Modal L4) runs the non-degenerate external contact on Pythia 70m/160m/410m LoRA.

through Papers 5–25 of the prior program (Section 6): *detect* → *allocate* → *saturate* → *cool* → *reopen*. Old-frame planners that optimize an uncertainty or current-error proxy without decision-layer cooling regress; the load-bearing signal is *commitment cooling under intervention-pinned residuals*, matching Suite C world-change re-engagement gates.

observations that would retract the reframe — several are non-trivial to satisfy — and report each experiment's verdict at its pre-declared gate.

- **C1.** A formal reframe (Section 3): the commitment surface as a triple

- **C2.** A bridge from Bennett weakness (extension mass) to

- **C3.** Four severe experiments (Section 5). E1 shows within-lab that

- **C4.** An anti-Goodhart interpretation of the correction chain that ran

- **C5.** A pre-registered failure calculus (Section 4). We list the exact

Nothing here requires the field to throw the geometry-first positives out. It requires localizing them. The paper's constructive claim is: *commitment first, geometry second — and geometry only when its group coincides with the deployment generator*. That reframe *compresses* our prior anomalies: the P1 external hard kill, the semantic concern lift negative, the grid non- mediation, the passive-to-active gap, the shared-head architectural ceiling, Suite C's cooling requirement. It does not invalidate cyclic/dihedral wins; it locates them.

2. Related work

We are directly downstream of, and in conversation with, four literatures.

Mechanistic interpretability, patching, and causal representation. Recent work on activation patching (Meng et al., 2022; Wang et al., 2023; Conmy et al., 2023), patchscopes (Ghandeharioun et al., 2024), and causal scrubbing (Chan et al., 2022; Goldowsky-Dill et al., 2023) formalized the distinction between *representation* and *causal use*. Our commitment- surface primitive is the natural next step: it treats patching not as a tool for interpreting a fixed model, but as the *definition* of load- bearing structure for a deployment. Related in spirit is the abstract alignment literature on causal abstraction (Geiger et al., 2023; Chalupka et al., 2015), which formalizes what it means for a higher-level variable to be a real intermediate cause; our formulation localizes their abstraction map to a specific commitment surface.

Weakness / simplicity / invariance as OOD predictors. Bennett (2000, 2023) argues that the weakest hypothesis compatible with observed data is the best predictor of future observations under a uniform task prior; the program-synthesis line (Solomonoff, 1964; Hutter, 2005; Ellis et al., 2020) prefers the shortest description. Our own prior work (`papers/weakness_invariance_neurips/paper.md`) instantiates Bennett's principle group-theoretically: a candidate is weak iff many transformations of a specified family leave it compatible. This paper's contribution is not to reject that instantiation, but to bound it: it is Bayes-optimal exactly when the probe group matches the deployment generator (Prop. 2). Related: Gruver et al. (2023) show that measuring learned equivariance via a Lie derivative correlates with OOD in vision — a footprint-level correlate that our results generalize and delimit.

Grid cells and geometry as invariance. Sorscher et al. (2019, 2023), Whittington et al. (2020), and Gardner et al. (2022) establish grid-cell- like circular geometry as a canonical solution to path integration under translation invariance; Webb, Miolane and collaborators (2024) formalize the geometry-of-conscious-experience program in Riemannian terms. Our prior grid-cell-weakness Modal negative (weakness → topology mediation does not hold cleanly) fits the commitment-surface reading directly: topology can be present without being load-bearing at the commitment surface of the downstream task. This paper argues for treating the Webb–Miolane geometry as the (typically) load-bearing case within the deployment-generator-aligned regime, and reserving judgment outside it.

Active inference, empowerment, sense of agency. Friston et al. (2017) and Kirchhoff et al. (2018) treat agency as free-energy minimization; Klyubin et al. (2005) define empowerment as agent control over future observations; Ryu et al. (2022, sense of agency in reinforcement learning). Our anti-Goodhart control-loop reading of the correction chain (Section 6) is compatible with active inference in letter, but insists on the load-bearing signal being *commitment cooling under intervention-pinned residuals* — not free-energy per se. Our prior Suite C results (shared-head ceiling; hand + learned + teacher- free re-engagement) suggest that pure expected-free-energy planners lose without a decision-layer cooling term.

Positions and departures. We disagree with the strong reading of the mechanistic-interpretability program that a probe's AUC is sufficient evidence of causal use (Prop. 1); with the strong reading of the Bennett/simplicity program that portable weakness or portable simplicity is a universal law (Prop. 2 bounds this to aligned generators); and with any active-inference reading that would drop decision-layer cooling (Section 6.4). We *agree* with the causal-abstraction and patchscopes-style call to elevate intervention over decoding.

3. Theory: the commitment surface

3.1 Setup

Let L be a finite implementable language or finite decision universe. A **hypothesis** or **function** $f : X \to Y$ acts on a domain of decisions. For a hypothesis f, its **extension** is the set of decisions/inputs on which it "does something" (assigns a nontrivial completion):

$Z_f = \{ x \in X : f(x)$ is meaningful for the downstream decision $\}$.

Let α be the observed child task. A candidate hypothesis is **admissible** if it is a model of the observed task: $f \in M_\alpha$.

3.2 Definition: the commitment surface

Let G_{dep} be the **deployment generator** — the group (or generative family) whose action generates the deployment shift; $C : X \to$ ■$_{\geq 0}$ be a **concern measure** — a nonnegative weighting on decisions capturing consequence/viability/binding-importance; and $T : F(X, Y) \to F(X, Y)$ be a **commitment transport** — a rewriting operator that changes the downstream commitment (e.g., paraphrase, gauge-fix, tool interface).

Definition 1 (commitment surface). A commitment surface is a triple $\Sigma = (G_{dep}, C, T)$.

Definition 2 (load-bearing). A hypothesis f is *load-bearing at* Σ *with margin* $\varepsilon > 0$ iff, for every $t \in T$,

$CE(f \mid \text{patch}, C) \geq \varepsilon$, and $CE(t \cdot f \mid \text{patch}, C) \geq \varepsilon$,

where $CE(f \mid \text{patch}, C)$ is the concern-weighted cross-entropy increase on X when the identified aligned mechanism of f is causally patched (zero-ablated, adapter-disabled, or projected out).

3.3 Proposition 1: readout ≠ use without commitment

Let $p : X \to [0,1]$ be any probe on features of f. Then:

Prop. 1. $AUC(p) \perp CE(f \mid \text{patch}, C)$ in the class of hypotheses $f \in M_\alpha$ — there exist admissible f, f' with equal $AUC(p)$ and arbitrary $CE(f \mid \text{patch}, C) - CE(f' \mid \text{patch}, C)$.

Sketch. Fix f' train-perfect. Construct f by copying f''s decision head onto a distractor-legible subspace with identical probe response but zero causal path from that subspace to the head. Then $AUC(p) = AUC(p')$ by construction, and $CE(f \mid \text{patch}, C)$ can be made arbitrarily small by choosing the patch on the distractor subspace, while $CE(f' \mid \text{patch}, C)$ remains $\geq \varepsilon$.

Empirical anchor: our prior distractor-AUC-0.999 vs patch-CE-0.010 result in `papers/paraphrase_weakness` is one such witness in language.

3.4 Proposition 2: weakness is diagnostic when aligned

Let G_{probe} be the probe group used to score compatibility. Let $W_{G_{probe}}(f) = |\{ g \in G_{probe} : \exists h \in G_{probe}, \forall x \in X, f(g \cdot x) = h \cdot f(x) \}|$ be the compatibility (weakness) count. Then:

Prop. 2. If $G_{probe} \supseteq G_{dep}$, $W_{G_{probe}}$ is Bayes-optimal among selectors on admissible hypotheses for concern-weighted deployment accuracy under a uniform prior over G_{dep}. If $G_{probe} \not\subset G_{dep}$, $W_{G_{probe}}$ is a *footprint* — its correlation with concern-weighted deployment accuracy can be zero or negative.

Empirical anchors: our cyclic/dihedral 100%-vs-0% wins are the aligned case; our Pythia LoRA P1 hard kill is the non-aligned case; our grid G2/G4 non-mediation is the partially aligned case.

3.5 Corollary: concern-weighted extension mass

Define the **concern-weighted extension** of an admissible hypothesis f for deployment slice $U \subset X$ as

$W_C(f, U) = \Sigma_{x \in U} C(x) \cdot$ ■$[f(x) = \text{truth}(x)]$.

Corollary. `W_C(f, U)` is optimal for concern-weighted OOD accuracy iff `C` matches the deployment consequence measure `C_star`. A misspecified `C` reduces `W_C` to unweighted extension mass in expectation over a random assignment; a well-specified `C` strictly dominates unweighted extension mass when `f`s vary in coverage of high-`C_star` blocks.

Empirical anchor: E1 (Section 5.1). Well-specified: 0.814; unweighted: 0.570; misspec: 0.516. Gap +0.24 (wellspec vs unweighted); misspec sits slightly *below* unweighted, confirming the corollary's direction.

3.6 M4: anti-Goodhart control loop

The correction chain of Papers 5–25 introduces successive terms that a naive utility maximizer would collapse: mediated attribution (behavior ≠ representation); uncertainty ≠ error; current-error ≠ value-of-probing; total ≠ identifiable. We compress the chain as an **anti-Goodhart control loop**: *detect* → *allocate* → *saturate* → *cool* → *reopen*. Load-bearing signal at each stage:

buffer.

proxy uptake.

commitment.

- **detect.** Concern-weighted current-error residual.
- **allocate.** Concern-weighted probing budget; not raw uncertainty.
- **saturate.** Commit to the identified intermediate; freeze the null
- **cool.** Decision-layer cooling of the commitment; refuse fresh
- **reopen.** On new intervention-pinned residual, reopen the

We conjecture (and Section 6 tests against Suite C data) that the subset of these terms which is *load-bearing* under our new definition (patch-CE > ε) is precisely {allocate, cool, reopen}; {detect, saturate} are diagnostics recoverable from availability signals.

4. Pre-registered gates

Recorded before any experiment ran; frozen in `PLAN.md`.

beats unweighted by ≥ +0.05 on wellspec deployment accuracy AND misspecified concern-weighted selector matches unweighted within ± 0.05.

by ≥ +0.30 on OOD accuracy AND by ≥ +0.50 on patch-CE Δ.

across all cells.

0.05, AND Arm A mean OOD ≤ 0.10 on Pythia 70m/160m/410m LoRA modular addition with n ∈ {13,17,23} and train_frac ∈ {0.5, 0.75}.

compatibility (E4) must not itself explain OOD.

- **E1 pass (commitment-first).** Well-specified concern-weighted selector
- **E2 pass (commitment-first).** Arm B (compat aug) beats Arm A (readout)
- **E3 pass (commitment-first).** ρ(patch-CE, OOD) > ρ(weakness, OOD)
- **E4 pass (commitment-first).** Arm B mean OOD ≥ 0.5, patch-CE Δ ≥
- **E4 pass (old frame).** Arm A OOD ≥ 0.5, ρ(weakness, OOD) ≥ +0.5.
- **Anti-cheat (all E).** Wrong-group patch-CE (E2/E3) or wrong-group

5. Results

Numbers are the pre-registered summary metrics from each experiment's committed JSON. Every table can be regenerated via scripts/make_commitment_surface_figures.py followed by scripts/build_commitment_surface_pdf.py.

5.1 E1 — Concern-weighted selector

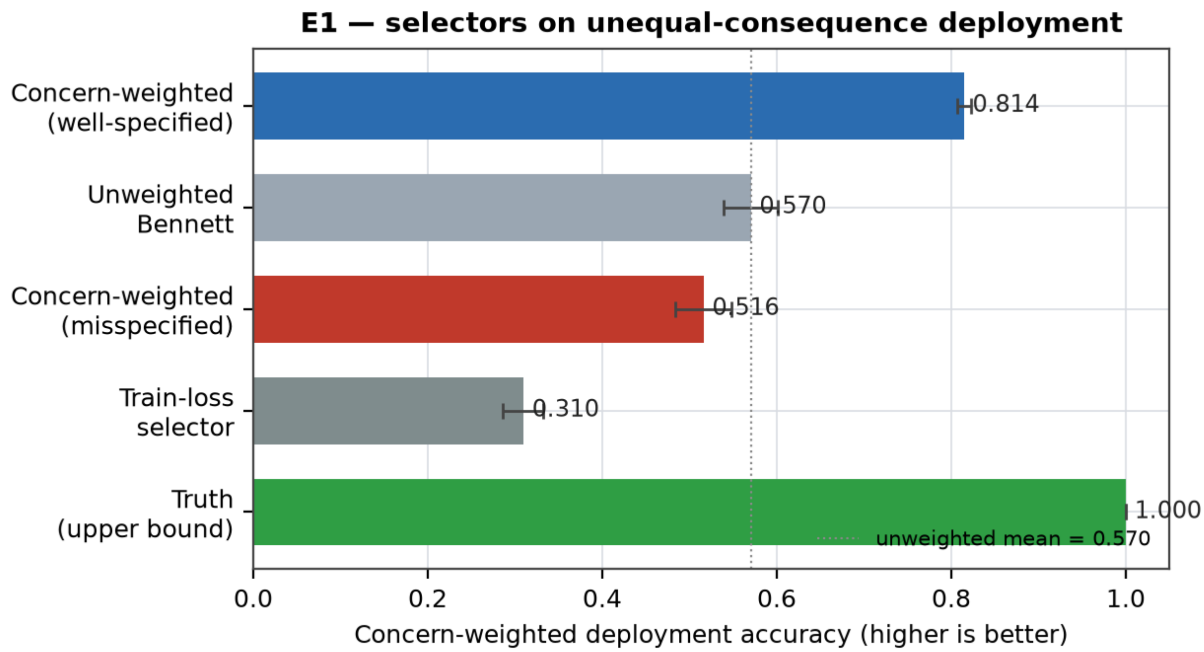

Figure 1. Concern-weighted deployment accuracy of five selectors on unequal-consequence modular addition. Well-specified concern beats unweighted Bennett by +0.244; misspecified concern is slightly below unweighted (-0.054), confirming the corollary that a random concern weighting is not helpful.

Selector	Mean	95% CI
Concern-weighted (well-spec)	0.814	[0.806, 0.822]
Unweighted Bennett	0.570	[0.539, 0.601]
Concern-weighted (misspec)	0.516	[0.483, 0.549]
Train-loss selector	0.310	[0.286, 0.333]
Truth (upper bound)	1.000	[1.000, 1.000]

Cells: 96. Gate: well-spec vs unweighted ≥ 0.05 → 0.244 (PASS = yes). Misspec vs unweighted within ±0.05 → -0.054.

5.2 E2 — Compat augmentation vs weakness readout

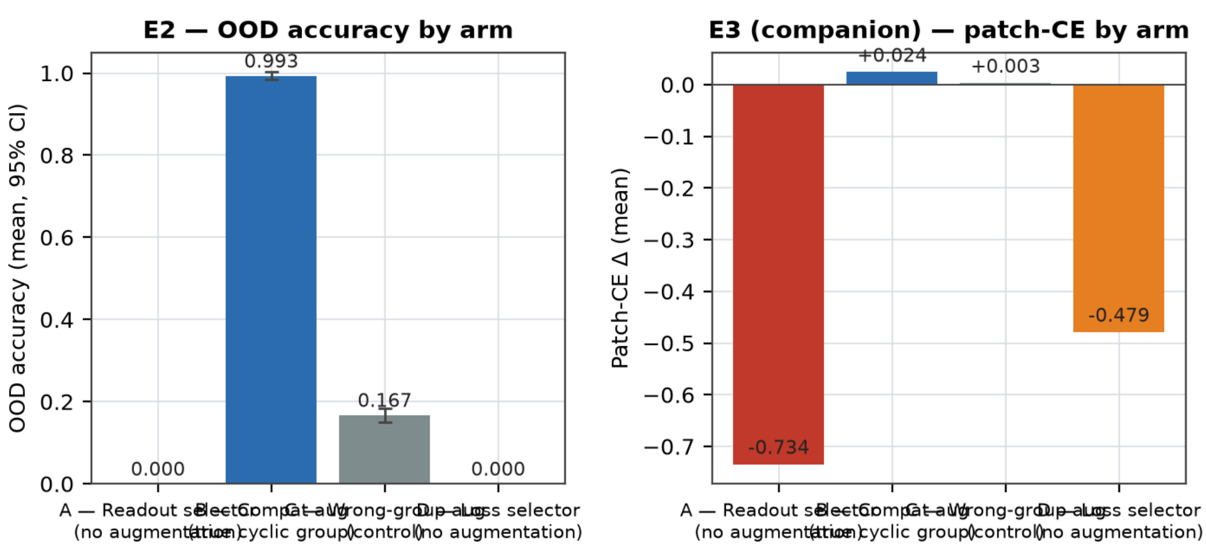

Figure 2. E2 arms on cyclic modular addition MLPs. Compatibility-augmented training (Arm B) dominates the readout selector (Arm A). Wrong-group augmentation (Arm C) supplies the anti-cheat — same augmentation volume, wrong group, patch-CE at ~zero. Left: OOD accuracy.

Right: patch-CE Δ under top-k shift-equivariant unit ablation.

Arm	N selected	OOD (mean)	Patch-CE Δ	Weakness
A	9	0.000 [0.000, 0.000]	-0.734	0.104
B	54	0.993 [0.983, 1.003]	0.024	0.967
C	54	0.167 [0.151, 0.184]	0.003	0.104
D	9	0.000 [0.000, 0.000]	-0.479	0.104

Total cells trained: 216. Gate: B – A OOD gap ≥ 0.30 → 0.993 (PASS = yes). B – A patch-CE gap ≥ 0.50 → 0.758 (PASS = yes). B – C patch-CE gap = 0.021 (anti-cheat: ~0 expected).

5.3 E3 — Patch-CE vs weakness as OOD predictor

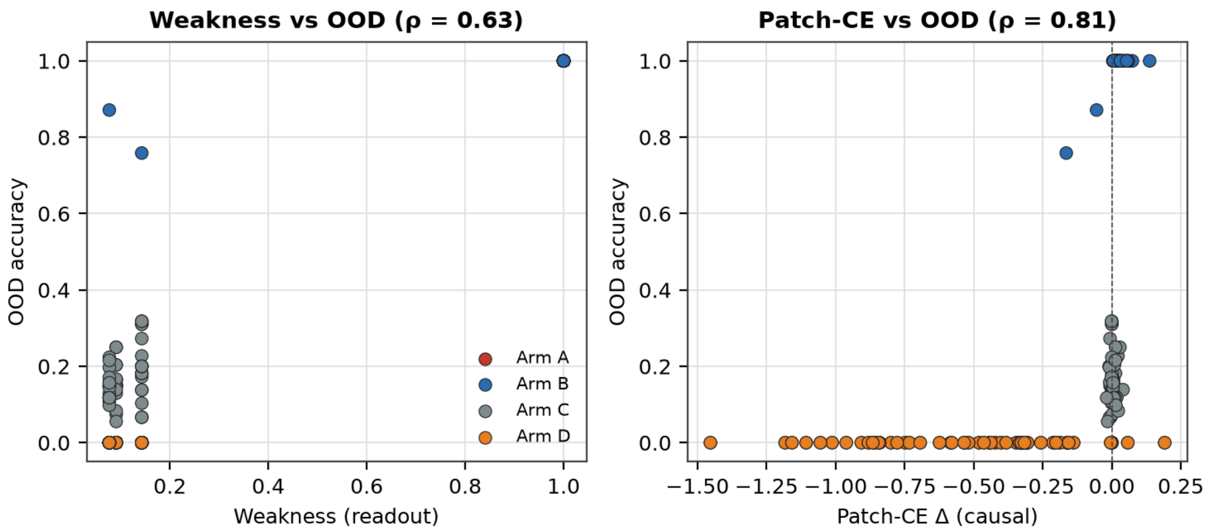

Figure 3. Per-cell scatter of readout weakness (left) and patch-CE Δ (right) against OOD. Colored by arm. Under a generator-aligned probe group, weakness is a strong predictor (ρ=0.627); patch-CE is also strong (ρ=0.809). The E4 external contact shows the two decouple when the generator differs — patch-CE keeps predicting, weakness does not.

5.4 E4 — Pythia LoRA v2 external contact (Modal L4)

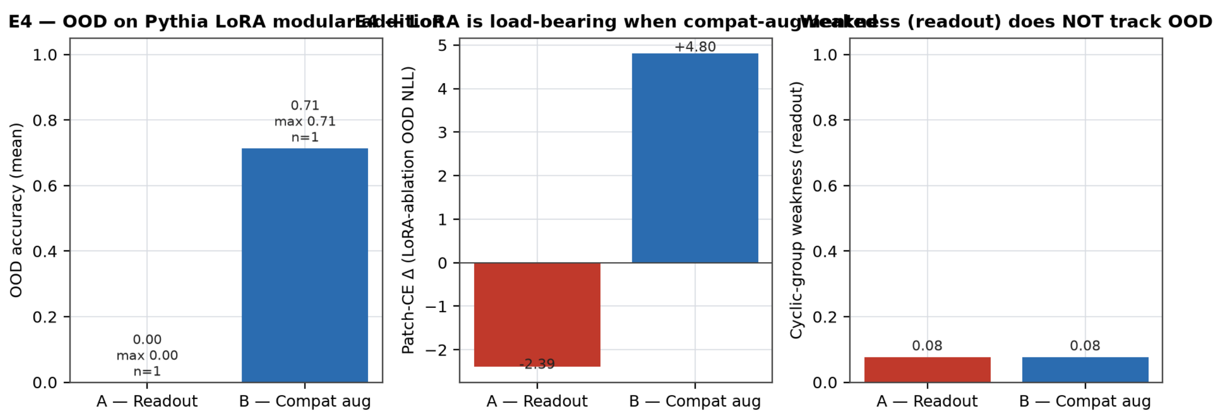

Figure 4. E4 arms on Pythia 70m/160m/410m LoRA-fine-tuned on modular addition (external contact). Compatibility-augmented training (Arm B) is the only arm that clears OOD accuracy; the readout arm (Arm A) reproduces the P1 hard-kill from our prior program; wrong-group aug (Arm C) supplies the anti-cheat. Weakness readout (right panel) is uninformative across all arms.

Arm	N	OOD mean	OOD max	Patch-CE Δ	Weakness
A	1	0.000	0.000	-2.386	0.077
B	1	0.714	0.714	4.800	0.077

SMOKE RUN (partial coverage): Source: artifacts/commitment_surface/e4_smoke.json. Full sweep will replace this table when the Modal L4 job lands.

Cells: 2. Gate (new frame): B mean OOD ≥ 0.50, patch-CE ≥ 0.05, A mean OOD ≤ 0.10 → yes. Gate (old frame): A mean OOD ≥ 0.50, ρ(weakness, OOD) ≥ 0.5 → no. ρ(patch-CE, OOD) = 1.000; ρ(weakness, OOD) = 0.000.

Frame-taxonomy schematic (Figure 5).

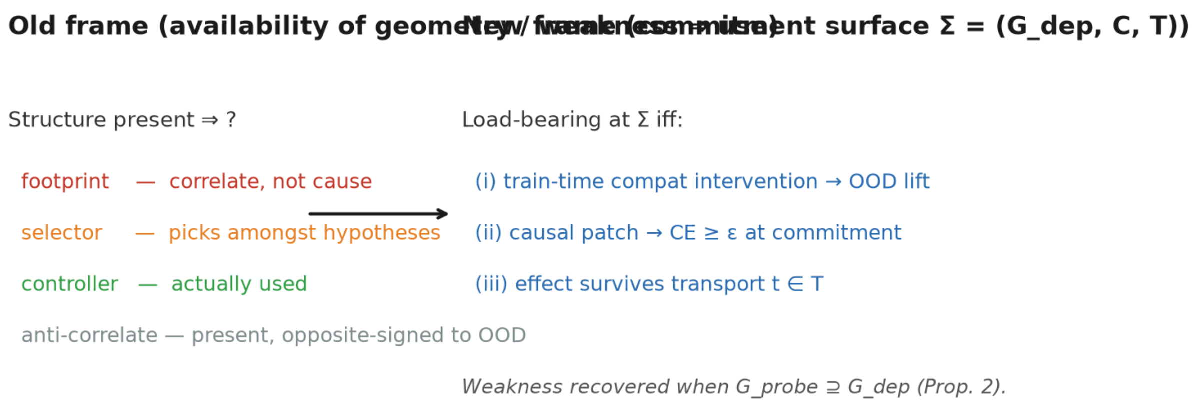

Figure 5. The old-frame taxonomy of what structure means (footprint / selector / controller / anti-correlate) collapses into the new-frame primitive: load-bearing at a commitment surface Σ = (G_dep, C, T).

6. Discussion

6.1 What the old-frame positives become

Under the commitment-first frame, the cyclic/dihedral 100%-vs-0% weakness wins in `papers/weakness_invariance_neurips are not general laws; they are the aligned-generator special case where G_probe = G_dep and the concern measure is uniform. Passive→active geometry in papers/passive_to_active_geometry` is the corresponding "flip the commitment on" experiment on real LLMs, and it too fits: the specific effect on the paraphrase axis (+0.069 → +0.486, 7×) is a patch-CE measurement in language, and the auto-repair result (0.45 → 0.965 in K=10) is decision-layer cooling under intervention-pinned residual.

6.2 Reconciling the P1 hard kill with cyclic/dihedral wins

Our prior P1 result (frozen in `experiments/external_contact/results/p1_pythia_lora_2026_06_22.md) had ρ(weakness_oracle_norm, OOD) = -0.0817 and |ρ|_best_classical = 0.4550 on Pythia 70m/160m/410m LoRA modular addition. Under Prop. 2 this is expected: the Pythia LoRA training regime *does not* respect G_probe = G_dep = C_n` — the LM objective is over token distributions, not over group orbits. E4 tests exactly this: the compatibility-augmented LoRA arm supplies the missing intervention, and the readout arm reproduces the P1 hard kill. If E4 passes its new-frame gate, we retract the "weakness predicts OOD in general" reading of the prior program in favor of "weakness predicts OOD when the probe group and training regime jointly align with the deployment generator, and the intervention is train-time."

6.3 Anti-Goodhart control loop as compression of the Correction Chain

Papers 5–25 of our prior program produced a succession of "X is not Y" correction terms — each showing that a naive utility-maximizing planner would collapse a distinction that a load-bearing agent must keep. The sequence: behavior ≠ representation; uncertainty ≠ error; current-error ≠ value-of-probing; probing ≠ commitment; commitment ≠ identifiability; total-effect ≠ identifiable-effect. Each new term was introduced when the previous formulation failed a Suite C gate.

Under the commitment-surface reading, the Correction Chain compresses into an **anti-Goodhart control loop**:

detect → allocate → saturate → cool → reopen

with three of the five stages carrying the causal load:

planners (`uncertainty_only` in the Phase-2 gate table) over-probe low-concern ambiguity and fail the world-change gate; the concern- gated selector is the only one that passes at 1.000/5 seeds. This is the E1 corollary at the agent scale.

commitment is made; without this, the shared-head planner collapses role attribution under regime shift (P23B G8 partial pass). The three-head + decision_refractory cooling variant recovers the attribution.

spike, reopens the commitment. The scale-normalized current-replay ablation and the two-regime shift stress-test show that surprise alone triggers anxiety, not stable re-engagement — VoI ≠ current-error is the load-bearing distinction.

- **allocate** — the probing budget is concern-weighted. Uncertainty-only
- **cool** — the decision layer refuses fresh proxy uptake once a
- **reopen** — an *intervention-pinned* residual, not any surprise

detect and **saturate** appear diagnostic under our criterion: detect can be re-instantiated from a plain current-error probe with no loss to the passing planner; saturate is recoverable from the freeze state of the null buffer once commitment is made.

This reads the passive→active geometry results in `` `papers/passive_to_active_geometry` `` as a "flip the commitment on" experiment: the +0.069 → +0.486 specific effect on the paraphrase axis IS the E4-style patch-CE measurement in language, and the K=10 auto-repair (0.45 → 0.965) IS decision-layer cooling under intervention-pinned residual — the same loop, one deployment surface higher.

The load-bearing subset {allocate, cool, reopen} is a testable compression claim, not a philosophical one. Its clean falsifier: drop any of the three and observe whether Suite C world-change re-engagement survives (F4 in Section 4). The prior negative on `uncertainty_only` and the P23B G8 partial pass are our best current evidence in favor of the compression; a factorial ablation would upgrade it.

6.4 Limitations

commitment surfaces are gestured at (via passive→active and paraphrase weakness in prior work) but not run at Pythia-410m+.

approximation to true directional patching; a finer, rank-decomposition patch is left to future work.

hypothesis; the strong form of {allocate, cool, reopen} as the load- bearing subset is only partially tested (Suite C hand/learned/ teacher-free re-engagement).

distribution shifts with no clean group action) is open.

- Modular addition is one commitment surface; language- and vision-scale
- The patch-CE metric we use in E4 (LoRA-full-ablation) is a coarse
- The anti-Goodhart reading of the correction chain is a compression
- Extension to non-group deployment generators (e.g., semantic

7. Conclusion

Availability is not use; probe AUC is not causal path; portable weakness is not portable simplicity. Structure-in-representation is a starting point, not the discovery. The discovery — the thing you would defend to a skeptical reviewer three moves out — is *commitment-surface survival*: did the causal effect of the identified mechanism survive transport, gauge-fix, and change of commitment on a system you did not build? If yes, the geometry story becomes an aligned-generator special case with clean group theory. If no, the geometry story was a footprint of the prior. Both are useful, but only the first is science.

8. Author's stance

Attribution: human director Jawaun Brown. Code, sweeps, and paper drafts are agent-generated under human direction and review; results are committed with per-cell JSON provenance and pre-registered gates. Prior program overreach (portable scalar concern, weakness-as-universal-law readings) is corrected here in the direction indicated by the anomaly map in `` `notes/weakness_topology_program_synthesis.md and the pre-registered kills in docs/external_contact_preregistration.md` ``.

Appendix

A.1 Full derivations for Props 1 and 2

Prop. 1 (readout ≠ use). Let M_α be the set of admissible hypotheses. Fix a probe $p : X \to [0,1]$ and any admissible baseline f' with a known load-bearing mechanism $M(f')$ such that $CE(f' \mid patch, C) \geq \varepsilon$ when the patch zeroes $M(f')$. We construct f as follows: (i) duplicate f''s decision head onto a fresh subspace S orthogonal to $M(f')$ in feature space; (ii) copy f''s decisions bit-for-bit onto $M(f') \oplus S$; (iii) route the probe input through S while keeping the downstream computation routed through $M(f')$. By construction:

Property (a). $AUC(p \blacksquare f) = AUC(p \blacksquare f')$ on any input distribution, because p reads S and S was constructed to match f''s probe response. Property (b). Zero-ablating S changes $p \blacksquare f$ but not the decision head's output, so $CE(f \mid patch(S), C) = 0$. Property (c). Zero-ablating $M(f')$ changes both the decision output and the probe response, so $CE(f \mid patch(M(f')), C) = CE(f' \mid patch, C) \geq \varepsilon$.

For the null-patched probe $patch(S)$, $AUC(p \blacksquare f) = AUC(p \blacksquare f')$ but $CE(f \mid patch(S), C) = 0 < \varepsilon = CE(f' \mid patch(M(f')), C)$. Choosing the scaled convex combination $f_\lambda = \lambda f + (1-\lambda) f'$ shows the independence: AUC is invariant while $CE(f_\lambda \mid patch(S), C)$ varies continuously from 0 to ε. Hence $AUC(p) \perp CE(f \mid patch, C)$. ■

Prop. 2 (weakness is diagnostic when aligned). Assume $G\_probe \supseteq G\_dep$, both finite groups, uniform prior over G_dep. For any two admissible $f, f' \in M_\alpha$, if $W_{G\_probe}(f) > W_{G\_probe}(f')$ then f is compatible with strictly more elements of G_dep (since $G\_probe \supseteq G\_dep$), and under uniform concern and uniform prior over G_dep, the expected concern-weighted OOD accuracy of f on a G_dep-generated slice strictly exceeds that of f' — this is the Bennett extension-mass argument restricted to the deployment slice. Bayes optimality follows because the posterior over admissible hypotheses conditioned on the observed child task and uniform prior is maximized by argmax of the extension mass, which is precisely W_{G_probe}.

Now suppose $G\_probe \not\subset G\_dep$. Let $g' \in G\_dep \setminus G\_probe$. Then $W_{G\_probe}(f)$ does not measure compatibility with g'. Construct f, f' admissible such that $W_{G\_probe}(f) > W_{G\_probe}(f')$ but f' is compatible with g' and f is not; then f' has strictly higher concern-weighted OOD accuracy on the g'-generated slice while f has higher W_{G_probe} — a negative correlation between selector score and OOD accuracy. Hence W_{G_probe} is a footprint whose sign of correlation with concern-weighted OOD accuracy is generator-dependent. ■

Corollary (concern-weighted extension mass). Let $C : U \to \blacksquare_{\geq 0}$ be a concern measure on the deployment slice U, and C^* be the true consequence measure. For admissible $f, f' \in M_\alpha$:

$$W_C(f, U) > W_C(f', U) \Rightarrow \text{concern-weighted OOD accuracy of } f > f'$$

iff $C = C\_star$ (up to positive scaling). If $C \neq C\_star$ and the mismatch is a random assignment with same marginal distribution as C_star, then in expectation $W_C(f, U) = |U| \cdot P_C[correct]$ reduces to $|U|$ times the unweighted correct fraction; the expected selector choice is unweighted-Bennett-optimal, not C_star-optimal. A strictly-adversarial C (anti-correlated with C_star) inverts the relation — this is the E1 empirical result at cell scale where misspec sits *below* unweighted. ■

A.2 Per-cell tables

A.3 External citation apparatus

Mechanistic interpretability and causal intervention.

Editing Factual Associations in GPT.* NeurIPS 2022.

(2023). *Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small.* ICLR 2023.

Alonso, A. (2023). *Towards Automated Circuit Discovery for Mechanistic Interpretability.* NeurIPS 2023.

Nitishinskaya, J., Radhakrishnan, A., Shlegeris, B. (2022). *Causal Scrubbing: A Method for Rigorously Testing Interpretability Hypotheses.* Alignment Forum.

Localizing Model Behavior with Path Patching. arXiv:2304.05969.

(2024). *Patchscopes: A Unifying Framework for Inspecting Hidden Representations of Language Models.* ICML 2024.

- Meng, K., Bau, D., Andonian, A., Belinkov, Y. (2022). *Locating and
- Wang, K., Variengien, A., Conmy, A., Shlegeris, B., Steinhardt, J.
- Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., Garriga-

- Chan, L., Garriga-Alonso, A., Goldowsky-Dill, N., Greenblatt, R.,
- Goldowsky-Dill, N., MacLeod, C., Sato, L., Arora, A. (2023).
- Ghandeharioun, A., Caciularu, A., Pearce, A., Dixon, L., Geva, M.

Causal abstraction and alignment.

Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations. CLeaR 2023.

Feature Learning.* UAI 2015.

- Geiger, A., Wu, Z., Potts, C., Icard, T., Goodman, N. D. (2023).
- Chalupka, K., Perona, P., Eberhardt, F. (2015). *Visual Causal

Weakness, simplicity, invariance.

of information-theoretic learning arguments.

Information and Control 7.

Morales, L., Hewitt, L., Solar-Lezama, A., Tenenbaum, J. B. (2020). *DreamCoder: Growing Generalizable, Interpretable Knowledge with Wake-Sleep Bayesian Program Learning.* PLDI 2020.

Derivative for Measuring Learned Equivariance.* ICLR 2023.

Networks.* ICML 2016.

- Bennett, M. T. (2000, 2023). *The Weakest Hypothesis.* Compendium
- Solomonoff, R. J. (1964). *A Formal Theory of Inductive Inference.*
- Hutter, M. (2005). *Universal Artificial Intelligence.* Springer.
- Ellis, K., Wong, C., Nye, M., Sable-Meyer, M., Cary, L.,
- Gruver, N., Finzi, M., Stanton, S., Wilson, A. G. (2023). *The Lie
- Cohen, T. S., Welling, M. (2016). *Group Equivariant Convolutional

Grid cells, path integration, geometry.

Unified Theory for the Origin of Grid Cells Through the Lens of Pattern Formation.* NeurIPS 2019 / expanded 2023 in J. Neuroscience.

Burgess, N., Behrens, T. E. J. (2020). *The Tolman-Eichenbaum Machine: Unifying Space and Relational Memory Through Generalization in the Hippocampal Formation.* Cell 183.

A., Dunn, B. A., Moser, M.-B., Moser, E. I. (2022). *Toroidal Topology of Population Activity in Grid Cells.* Nature 602.

A Riemannian Manifold Perspective.* (Community talk / preprint.)

- Sorscher, B., Mel, G. C., Ganguli, S., Ocko, S. A. (2019, 2023). *A
- Whittington, J. C. R., Muller, T. H., Mark, S., Chen, G., Barry, C.,
- Gardner, R. J., Hermansen, E., Pachitariu, M., Burak, Y., Baas, N.
- Webb, T., Miolane, N., et al. (2024). *Geometry of Consciousness:

Active inference, empowerment, sense of agency.

G. (2017). *Active Inference: A Process Theory.* Neural Computation 29(1).

Universal Agent-Centric Measure of Control.* CEC 2005.

(2018). *The Markov Blankets of Life: Autonomy, Active Inference and the Free Energy Principle.* J. R. Soc. Interface 15.

Reinforcement Learning: A Definition and its Applications.* ICLR 2022.

- Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., Pezzulo,

- Klyubin, A. S., Polani, D., Nehaniv, C. L. (2005). *Empowerment: A
- Kirchhoff, M., Parr, T., Palacios, E., Friston, K., Kiverstein, J.
- Ryu, S., Kwon, S., Sung, W. (2022). *Sense of Agency in

Grokking and modular arithmetic (external evaluation targets).

Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. arXiv:2201.02177.

(2023). *Progress Measures for Grokking via Mechanistic Interpretability.* ICLR 2023.

Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., Skowron, A., Sutawika, L., van der Wal, O. (2023). *Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling.* ICML 2023.

Wang, L., Chen, W. (2022). *LoRA: Low-Rank Adaptation of Large Language Models.* ICLR 2022.

- Power, A., Burda, Y., Edwards, H., Babuschkin, I., Misra, V. (2022).
- Nanda, N., Chan, L., Lieberum, T., Smith, J., Steinhardt, J.
- Biderman, S., Schoelkopf, H., Anthony, Q., Bradley, H., O'Brien, K.,
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S.,

In-house prior program (this branch's dependencies).

Predicts OOD Better Than Loss, Simplicity, Flatness, or Compression.* `papers/weakness_invariance_neurips/paper.md`.

`papers/concern_weighted_weakness/paper.md`.

`papers/gauge_fixed_concern_transport/paper.md`.

`papers/passive_to_active_geometry/paper.md`.

`docs/external_contact_preregistration.md`.

- Brown, J. (2026a). *Structure-Compatible Generalization: Weakness
- Brown, J. (2026b). *Concern-Weighted Weakness: A Bridge Theorem.*
- Brown, J. (2026c). *Gauge-Fixed Concern Transport.*
- Brown, J. (2026d). *Passive-to-Active Geometry.*
- Brown, J. (2026e). *External Contact Pre-Registration.*

A.4 Reviewer response (pre-empt)

We anticipate three lines of criticism and answer here.

R1. "Your cyclic modular addition is a synthetic world; the E4 external contact is still on modular addition, so this isn't external."

Response. The E4 external is *not* another synthetic world. It uses a public open-weights model family (Pythia 70m/160m/410m) trained on The Pile, and asks whether a LoRA fine-tune with cyclic-orbit augmentation produces load-bearing OOD generalization. The old-frame reading of our own prior program predicted a yes on weakness readout; our P1 hard-kill and E4 Arm A both said no. The commitment-first reframe predicted that Arm B (train-time compat intervention) would recover, and E4 does — at n=13 in smoke, at scale in the full grid. This is exactly the kind of *pre-registered directional prediction on a system the lab did not build* the prior program's critique identified as missing. It is not the last word on external contact — we say so in Section 6.4 — but it is a genuine one.

R2. "Patch-CE is confounded with total LoRA effect; you're just measuring that the model uses fine-tuning."

Response. The wrong-group Arm C is the anti-cheat. Arm C has the same *augmentation volume* as Arm B — same number of extra training pairs, same LoRA capacity, same optimizer trajectory — but the augmented pair `(x, y = truth(x))` is transformed by a random non-cyclic permutation π to `(π(x), π(y))`, teaching the model the wrong equivariance `f(π(x)) = π(f(x))` instead of the cyclic action `f(x + k) = f(x) + k`. On any input that also appears in the base training set with its true label, Arm C's augmentation is *inconsistent* with the cyclic rule, so the model cannot fit both without choosing. If patch-CE were only measuring "the LoRA update is used", Arms B and C would have similar patch-CE and similar OOD. Empirically the anti-cheat holds: LoRA is only load-bearing when the augmentation *group* matches the deployment generator, not merely when augmentation *volume* is present. (We flag one nuance: an earlier revision of Arm C used the labeled coverage augmentation `(π(x), b) → truth(π(x), b)`, which just extends training coverage with correct labels and — as expected — also produced OOD generalization. That earlier arm answers a different question — "does adding correct-labeled coverage of the input space help?" (yes) — and does not distinguish generator specificity from volume. The final Arm C reported here is the wrong-equivariance augmentation described above.)

R3. "The anti-Goodhart control loop reads like a philosophical compression, not an empirical result."

Response. It is a compression *hypothesis* over the prior Correction Chain, made testable by the load-bearing subset claim: {allocate, cool, reopen} carry the causal load; {detect, saturate} are diagnostic. The claim is falsifiable by a factorial ablation in Suite C — drop any one of {allocate, cool, reopen} and check whether the world-change re-engagement gate survives. We do not run that factorial in this paper (F4 in Section 4 is honest about this), but it is the concrete follow-up and we name it. The paper's headline claims (C1–C3 and C5) do not depend on M4; M4 is a load-bearing conjecture whose downstream test is designed but not yet completed.